

Chapter 2

Properties of Earth Materials

Most physical properties of rocks are not constant, but change as state conditions change (pressure and temperature). Composition is an important control, as are the crystallographic structure, crystal orientation, crystal defects, porosity, and grain size. There are several methods that one may use to estimate physical properties including by oxide composition, mineralogy, and thermodynamic modeling. Each have their own drawbacks and advantages.

Oxide models often employ simple empirical functions that make them easy to apply, but are generally the least flexible and may extrapolate poorly beyond their calibrated range of compositions. Mineralogic models may be calibrated to a range of laboratory pressure and temperature conditions and can incorporate crystal orientation and/or compositional layers, but there are no guarantees that the mineral assemblage is thermodynamically stable. Thermodynamic-based computations are time intensive to compute but cannot incorporate geometric considerations.

So which should one choose? It depends on how accurate one must be for the desired application.

In this chapter, you will find a set of formulae and tables of empirically determined laboratory constants that can be used to estimate the physical properties of minerals and mixing models that can be used to estimate the bulk physical properties of rocks built from these minerals. The methods described in this chapter generally pertain to igneous and metamorphic rocks without the presence of a fluid phase, although fluids could be included in a similar framework. Near the surface we often have to contend with porosity and/or cracks that add additional controls on the physical properties that are not easily handled by this framework.

More on each physical property is included within the individual chapters pertaining to the fields that are sensitive to them.

2.1 P & T effects on mineral properties

2.1.1 Density, ρ [kg m^{-3}]

Density is the ratio of a unit mass to a specified volume. The density at 298 K is determined by the formula weight per molar volume, i.e., $\rho^\circ = m_{\text{fu}}(V^\circ)^{-1}$. However, very little of the Earth is at 298 K (25°C) and atmospheric pressure (0.1 MPa). Rock density changes in response to the higher temperatures and pressures found internal to the Earth. Therefore, in order to appropriately model density, we must, in many cases, account for the difference with respect to surface conditions.

To estimate density changes with respect to temperature only,

$$\rho(T) = \rho^\circ e^{-A}, \quad (2.1)$$

where ρ° is the density at reference temperature and pressure (0.1 MPa and 298 K), and T° is the reference temperature, and A is the integral of expansivity (Equation ??). To compute the density change with pressure and temperature we can use the approximation

$$\rho(T, P) = \rho(T) \left(1 + \frac{P - P^\circ}{K_T(P, T)} \right). \quad (2.2)$$

This approximation is most applicable within the crust and uppermost mantle (see Bulk Modulus below).

2.1.2 Expansivity, α [K^{-1}]

Expansivity also known as the coefficient of thermal expansion, represents the change in volume in response to changes in temperature.

The temperature-dependent volumetric expansivity, α , is estimated by

$$\alpha(T) = a_0 + a_1 T + a_2 T^{-2}, \quad (2.3)$$

where a_i are empirical constants (?). The P - and T -dependent thermal expansion is computed by

$$\alpha(P, T) = \alpha(T) \left(\frac{\rho(P)}{\rho^\circ} \right)^{-\delta_T}, \quad (2.4)$$

where ρ° and $\rho(P)$ are the densities at 0 GPa and at the pressure P , and δ_T is the Anderson-Grünison parameter.

Note that in order to compute the pressure dependence of expansivity, you will need the isothermal bulk modulus corrected for pressure. This would ordinarily require an iterative or root finding method, but we will simplify it for our purposes since the error introduced with our approximations are small enough for our purposes.

To estimate several parameters, we need the integral of the thermal expansion with respect to temperature, i.e.,

$$A = \int_{T^\circ}^T \alpha(T) dT. \quad (2.5)$$

Since our expression for thermal expansion is relatively simple,

$$A = a_0(T - T^\circ) + \frac{a_1}{2}(T^2 - (T^\circ)^2) - a_2(T^{-1} - (T^\circ)^{-1}), \quad (2.6)$$

where T° is the reference temperature, which is defined as 298 K.

Likewise, the derivative of expansivity is necessary to estimates some parameters,

$$\frac{\partial \alpha}{\partial T} = a_1 - 2a_2 T^{-3}. \quad (2.7)$$

2.1.3 Bulk Modulus, K_T & K_S [GPa]

The bulk modulus is the inverse of compressibility, that is a measure of the change in volume in 3-dimensions in response to a change in pressure. The bulk modulus is defined in two ways depending upon the whether the volume change occurs at constant temperature (isothermal) or at constant entropy (isentropic).

We can approximate the change in isothermal bulk modulus with respect to temperature as

$$K_T(T) = K_T^\circ e^{-\delta_T A}, \quad (2.8)$$

and with pressure, we can approximate the bulk modulus as

$$K_T(T, P) = K_T(T, P^\circ) + K'_T(P - P^\circ), \quad (2.9)$$

where P° is atmospheric pressure. The pressure dependence above is overly simplistic for higher pressures, where the Birch-Murnaghan equation of state is used as a more accurate approximation.

The isentropic bulk modulus, K_S , is related to the isothermal bulk modulus, K_T , by

$$K_S = K_T(P, T)[1 + T\gamma_{th}\alpha(P, T)]. \quad (2.10)$$

2.1.4 Shear Modulus, μ [GPa]

The shear modulus is a measure of the resistance of a material to deform under shear. The temperature response to the shear modulus is given by

$$\mu(T) = \mu^\circ e^{-\Gamma A}, \quad (2.11)$$

where $\Gamma = \left(\frac{\partial \ln \mu}{\partial \ln \rho}\right)_P$ (see Table ??). Just as the bulk modulus, we can estimate the shear modulus with a pressure effect as

$$\mu(T, P) = \mu(T, P^\circ) + \mu'(P - P^\circ). \quad (2.12)$$

2.1.5 Seismic Velocity, V_P & V_S [km s⁻¹]

Seismic velocity is the speed at which sound travels through a medium. There are several different types of seismic waves. Two that we'll discuss most often. The compressional wave velocity is generally referred to as V_P , and the shear wave velocity is generally written V_S . Seismic velocities are derived quantities, dependent upon the bulk and shear moduli, and density. Once these parameters have been adjusted for pressure and temperature,

$$V_P = \left(\frac{K_S + \frac{4}{3}\mu}{\rho} \right)^{1/2} \quad (2.13)$$

$$V_S = \left(\frac{\mu}{\rho} \right)^{1/2}. \quad (2.14)$$

2.1.6 Specific Heat Capacity, C_P [J kg⁻¹ K⁻¹]

Specific heat capacity is the amount of additional heat required to raise a unit mass one kelvin. The specific heat capacity can be approximated as

$$C_P(P, T) = c_0 + c_1 T^{-0.5} + c_2 T^{-2} + c_3 T^{-3} + \frac{dC_P}{dP}(P - P^\circ), \quad (2.15)$$

with empirically derived constants, c_i (?). The change in specific heat capacity with respect to temperature is significantly greater than the change with respect to pressure. Hence we can often ignore the pressure effects. The pressure contribution is computed using the thermodynamic

identity (?),

$$\left(\frac{\partial C_P}{\partial P}\right)_T = -\frac{T}{\rho(P, T)} \left(\alpha^2 + \frac{\partial \alpha}{\partial T}\right)_P. \quad (2.16)$$

The bulk heat capacity of the rock matrix is computed as the mean heat capacity of all phases weighted by their volume fractions.

2.1.7 Thermal Conductivity, k [$\text{W m}^{-1} \text{K}^{-1}$]

Thermal conductivity is the rate of heat transfer per unit length per degree and represents the efficiency with which heat is transferred. There are two types of thermal conductivity: lattice and radiative. Radiative conductivity is only relevant at high temperatures and may not be significant compared to lattice conductivity so we'll just focus on the latter. The lattice conductivity for a given mineral is estimated by

$$k(P, T) = k^\circ \left(\frac{298}{T}\right)^n \left(1 + \frac{K'_T}{K_T}(P - P^\circ)\right). \quad (2.17)$$

Since most minerals are solid solutions of multiple end-members, the conductivity estimated between two end members can often be approximated by

$$k(P, T) = (k_0 + k_1 X + k_2 X^2) \left(\frac{298}{T}\right)^n \left(1 + \frac{K'_T}{K_T}(P - P^\circ)\right), \quad (2.18)$$

2.1.8 Thermal Diffusivity, k [$\text{mm}^2 \text{s}^{-1}$]

Thermal diffusivity is the rate at which heat is transferred. Computing thermal diffusivity is relatively simple once the other properties have been estimated because it is a derivative quantity. That is to say that it is determined from other properties, in this case

$$\kappa = \frac{k}{\rho C_P}. \quad (2.19)$$

2.2 Mixing Formulae

Most geologic materials are mixtures of either multiple minerals and/or fluids. However, when we make geological observations and model the fields we find that it is not possible to resolve these individual components. Instead, we are sensitive to the 'average' or 'effective' properties creating or modifying geophysical fields. Average is a bit of a dangerous word because it can indicate different things in different situations. So when someone says the computed the average always I want you to ask, "What kind?"

The typical person refers to the arithmetic mean when they use the term average. However, there are multiple types of averages that are mathematically defined. There is a mean, median and mode that you may remember from statistics. There are also an arithmetic, geometric, and harmonic means. Means can be weighted or unweighted. This section focuses on each of these averages and when they may be appropriately applied to Earth's materials.

2.2.1 Pythagorean means

Arithmetic

$$\text{unweighted:} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (2.20)$$

$$\text{weighted:} \quad \bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2.21)$$

$$(2.22)$$

Use: computing effective density, specific heat capacity

Geometric

$$\text{unweighted:} \quad \bar{x} = \left(\prod_{i=1}^n x_i \right)^{1/n} \quad (2.23)$$

$$\text{weighted:} \quad \bar{x} = \left(\prod_{i=1}^n x_i^{w_i} \right)^{1/\sum_{i=1}^n w_i} \quad (2.24)$$

$$(2.25)$$

Use: effective thermal conductivity for randomly oriented mixtures

The arithmetic and geometric means are mathematically related. Suppose we have a set of numbers, $x = (x_1, x_2, \dots, x_n)$. We can define a transform of $x \mapsto y$ by $y = \log x$. Arithmetic means are often appropriate when the values of something are often about the same order of magnitude, but geometric means make a bit more sense when the range of a property may span orders of magnitude as the do for many types of conductivity (e.g., electrical and hydraulic conductivity).

Harmonic

$$\text{unweighted:} \quad \bar{x} = \left(\sum_{i=1}^n x_i^{-1} \right)^{-1} \quad (2.26)$$

$$\text{weighted:} \quad \bar{x} = \left(\sum_{i=1}^n w_i \right) \left(\sum_{i=1}^n w_i x_i^{-1} \right)^{-1} \quad (2.27)$$

Use: effective thermal conductivity for layered conductivity packages

2.2.2 Additional mixing formulae

Voigt-Reuss-Hill average

The Voigt-Reuss-Hill average is really an average of the volume weighted arithmetic and harmonic means and is common for estimating the bulk or effective magnitude of a physical property. The average is given by

$$\bar{\Psi} = \frac{1}{2}(\Psi_V + \Psi_R) \quad (2.28)$$

$$= \frac{1}{2} \left[\sum_{i=1}^n \Psi_i v_i + \left(\sum_{i=1}^n \frac{v_i}{\Psi_i} \right)^{-1} \right], \quad (2.29)$$

where n is the number of constituent minerals and v_i is the volume fraction of the i -th component.

Use: moduli

Hashin-Strikman bounds

The Hashin-Shtrikman bounds are commonly quoted as a mixing relation for electrical conductivity and occasionally seismic velocity; however, the bounds often over- or under-predict the conductivity, but that is what they are designed to do. The bounds are

$$\sigma_{\text{HS}}^- = \mathcal{S}(\sigma_{\min}) \leq \sigma \leq \mathcal{S}(\sigma_{\max}) = \sigma_{\text{HS}}^+, \quad (2.30)$$

where $\mathcal{S}$ is defined as

$$\mathcal{S}(x) = \left(\sum_{i=1}^N \frac{\phi_i}{\sigma_i + 2x} \right)^{-1} - 2x, \quad (2.31)$$

and ϕ_i is the volume fraction of the i -th component.

Use: moduli, electrical conductivity

Table 2.1: Empirically derived constants used to compute physical properties of rocks.

mineral	formula unit	formula weight	molar volume*	density	H ₂ O wt%	expansivity			specific heat capacity				isothermal bulk modulus	$\frac{dK_T}{dP}$	shear modulus	$\frac{d \ln \mu}{d \ln \rho}$	$\frac{d\mu}{dP}$	Grüneisen parameter	Anderson-Grüneisen parameter
		(g mol ⁻¹)	(cm ³ mol ⁻¹)	@298 K (kg m ⁻³)		$\alpha = a_0 + a_1 T + a_2 T^{-2}$ (K ⁻¹)	$C_P = c_0 + c_1 T^{-0.5} + c_2 T^{-2} + c_3 T^{-3}$ (kJ kg ⁻¹ K ⁻¹)						(GPa)		(GPa)	Γ	μ'	γ_{th}	δ_T
		m_{fU}	V°	ρ°		$a_0 \times 10^5$	$a_1 \times 10^8$	a_2	c_0	c_1	c_2	c_3	K_T°	K'_T	μ°				
a-quartz	SiO ₂	60.1	22.7	2648	0	1.417	9.6581	-1.6973	1.7699	-23.49	45.947	-7.2006	37.12	5.99	44.80	40.1	0.46	0.7	8.42
b-quartz	SiO ₂	60.1	22.7	2530	0	-0.44	0	0	0.9647	-0.0004	30.546	-0.0018	57.01	4.00	41.40	4.1	1.21	0.1	4.11
orthoclase	KAlSi ₃ O ₈	278.3	105.1	2555	0	3.4	0	0	1.4445	-9.5303	-26.395	3.6781	58.30	4.00	28.10	4.4	0.80	0.4	4.44
albite	NaAlSi ₃ O ₈	262.2	99.0	2620	0	1.9801	1.0065	-0.976	1.5012	-9.2116	-30.099	4.0829	57.76	4.00	29.78	13.0	1.21	0.6	4.57
anorthite	CaAl ₂ Si ₂ O ₈	278.2	105.0	2760	0	1.2491	-0.0162	0.0161	1.5793	-4.7252	0	-1.1395	81.86	4.00	36.05	6.8	3.48	0.5	4.47
muscovite	KAl ₃ Si ₃ O ₁₀ (OH) ₂	398.2	150.4	2828	4.5	3.537	0	0	1.668	-10.994	-36.79	5.0614	49.00	4.00	29.54	4.5	1.00	0.5	4.52
phlogopite	KMg ₃ AlSi ₃ O ₁₀ (F;OH) ₂	417.2	157.6	2788	4.5	5.8	0	0	1.4354	-4.0508	-61.28	8.995	49.70	8.59	22.94	9.1	0.77	0.6	9.14
annite	KFe ₃ AlSi ₃ O ₁₀ (F;OH) ₂	511.9	193.3	3317	3.5	3.7	0	0	1.2777	-5.661	-42.813	6.9201	49.70	8.59	22.94	9.1	0.77	0.6	9.14
hornblende	Ca ₂ (Mg,Fe,Al) ₅ (Al,Si) ₈ O ₂₂ (OH) ₂	864.7	326.6	3248	2.5	2.075	1.027	0	1.6559	-13.126	1.5163	-2.6059	93.97	4.00	54.50	5.1	0.97	1.1	5.10
diopside	CaMgSi ₂ O ₆	216.6	81.8	3272	0	3.33	0	0	1.4103	-7.4109	-33.09	4.2567	113.00	4.80	64.90	6.0	2.00	1.2	6.04
hedenbergite	CaFeSi ₂ O ₆	248.1	93.7	3651	0	2.98	0	0	1.3712	-10.668	-32.729	-0.0042	119.20	4.00	61.00	5.2	0.85	1.2	5.21
enstatite	MgSiO ₃	200.8	75.8	3206	0	2.972	0.5711	0	1.6592	-11.959	-22.616	2.7805	105.80	8.50	76.80	9.4	2.00	0.9	9.39
ferrosilite	FeSiO ₃	263.9	99.6	4003	0	2.875	0	0	1.3204	-10.559	-3.444	-0.2858	100.17	4.00	52.00	5.1	2.00	1.1	5.05
forsterite	Mg ₂ SiO ₄	140.7	53.1	3222	0	2.854	1.008	-0.3842	1.6572	-12.804	0	-1.9042	127.30	5.37	81.60	5.2	1.82	1.2	5.50
fayalite	Fe ₂ SiO ₄	203.8	77.0	4400	0	2.386	1.153	-0.0518	1.2366	-9.882	0	-0.3052	136.70	5.16	51.00	4.7	0.62	1.1	5.40
spinel	MgAl ₂ O ₄	142.3	53.7	3575	0	1.96	1.64	0	1.7197	-14.086	0	0	196.20	4.00	107.81	4.2	0.92	1.3	6.50
pyrope	Mg ₃ Al ₂ Si ₃ O ₁₂	403.1	152.2	3565	0	2.311	0.5956	-0.4538	1.4657	-7.012	-33.042	3.1262	173.81	5.00	94.00	4.1	1.50	1.3	5.30
almandine	Fe ₃ Al ₂ Si ₃ O ₁₂	497.7	188.0	4324	0	1.776	1.214	-0.5071	1.2485	-6.605	-30.298	4.4437	174.10	6.00	92.10	5.5	1.60	1.0	5.52

*The ° indicates conditions at 0.1 MPa and 298 K.

Expansivity constants and specific heat capacity from ? and references therein. All other values from ? and references therein.

Table 2.2: Coefficients to estimate thermal conductivity

mineral	mole fraction of end-member	thermal conductivity			
		$k(P, T) = (k_0 + k_1 X + k_2 X^2) \left(\frac{298}{T} \right)^n \left(1 + P \frac{K'_T}{K_T} \right)$ (W m ⁻¹ K ⁻¹)			
	X	k_0	k_1	k_2	n
a-quartz		8.79	0	0	1.48
b-quartz		0.993	0	0	-1
orthoclase		1.791	0	0	-0.27
plagioclase	anorthite	2.199	-2.18	1.9	-0.21
muscovite		2.8	0	0	1.539
biotite		2.267	0	0	1.539
hornblende		2.6485	0	0	0.5
clinopyroxene		4.25	0	0	0.54
orthopyroxene		3.37	0	0	0.3
olivine	forsterite	3.091	-1.17	3.35	0.49
spinel		12	0	0	1.24
garnet	pyrope	4.966	-7.42	8.45	0.37

All values from ? and references therein.